

SONIA CHARLES THALATOTI

+1 (810) 333-0876 | thalatot@msu.edu | linkedin.com/in/sonia-charles-thalatoti | github.com/SoniaCharles

PROFESSIONAL SUMMARY

Software Engineer and M.S. Computer Science student at Michigan State University with experience building scalable web applications, REST APIs, cloud-deployed services, and AI-powered systems. Skilled in Python, C#, .NET, JavaScript, React, SQL, and Azure with hands-on experience delivering production-ready solutions through Agile development practices. Passionate about solving complex engineering problems across backend systems, distributed computing, and machine learning.

EDUCATION

Michigan State University	East Lansing, MI
Master of Science in Computer Science	GPA: 3.75/4.0 Aug 2025 – May 2026
Bachelor of Science in Computer Science and Information Systems [Dean's List]	GPA: 3.84/4.0 Aug 2021 – May 2025

TECHNICAL SKILLS

Languages: Python, C#, Java, JavaScript, SQL
Frontend: ReactJS, HTML5, CSS3, TypeScript
Backend: .NET, ASP.NET Core, Node.js, REST APIs, Entity Framework Core, MVC
Cloud & DevOps: Microsoft Azure, AWS S3, Git, GitLab CI/CD
Databases: SQL Server, MySQL
Tools: Postman, Visual Studio, SSMS

EXPERIENCE

Software Engineer Intern — TechClub Inc (<i>Remote</i>), Lincolnshire, IL	Sep 2025 – Dec 2025
- Designed and integrated RESTful APIs and webhook-based workflows to synchronize candidate data across AI analytics platforms and enterprise ATS systems. - Developed and debugged backend services using Node.js, improving reliability of real-time data exchange and automated recruiting workflows. - Collaborated with cross-functional stakeholders in Agile environments to translate business requirements into scalable technical solutions. - Built unit and integration tests that improved API stability, regression coverage, and long-term maintainability of production services.	
Backend Developer — Meijer Capstone Project, East Lansing, MI	Aug 2024 – Dec 2024
- Led backend development in C#/.NET and ASP.NET Core for a full-stack e-commerce simulation, implementing RESTful APIs for cart, checkout, and rewards functionality using Entity Framework Core and SQL Server. - Designed and optimized SQL Server schemas, stored procedures, LINQ queries, and Entity Framework Core data models to support high-performance transactional consistency. - Developed and tested backend services using Visual Studio, SSMS, and Postman; deployed applications on Microsoft Azure App Service with GitLab CI/CD and collaborated with frontend developers for ReactJS integration.	

PROJECTS

LLM-Assisted Simulation Verification	Python Formal Verification Ollama Gemma 3
- Developed a hybrid formal verification framework combining LLMs with verifier-guided refinement for finite-state transition systems. - Built automated validation pipelines for state mappings, transition consistency, and relation correctness. - Implemented counterexample-guided refinement, improving verification success rates from 0% to 100%.	
Yelp Review Sentiment Classification	Python Machine Learning NLP Scikit-learn
- Built multi-class sentiment classification pipelines on 10,000 Yelp reviews using SVM, Naive Bayes, Gaussian MAP, and Random Forest models. - Implemented TF-IDF, feature selection, and dimensionality reduction, achieving 58.39% cross-validation accuracy with SVM.	
Massively-Parallel Mandelbrot + GPU Microbenchmarks	C++ CUDA PTX Pthreads SIMD
- Implemented Mandelbrot rendering using SIMD vectorization, Pthreads parallelism, and CUDA GPU acceleration. - Wrote CUDA/PTX microbenchmarks analyzing FFMA throughput and memory latency across coalesced and non-coalesced access patterns.	
Cloud-Based Customer Data Pipeline	Python SQL AWS S3 PySpark Git
- Developed an end-to-end ETL pipeline using Python, SQL, Pandas, PySpark, and AWS S3 to clean, transform, and standardize customer transaction data. - Designed dimensional data models and optimized SQL queries to support scalable reporting, analytics, and downstream machine learning workflows.	

CERTIFICATIONS

Microsoft Applied Skills: Create an AI Agent	Microsoft, Apr 2026
Data Engineering on AWS - Foundations	Amazon Web Services (AWS), Apr 2026
Microsoft Certified Azure AI Engineer Associate (AI-102)	Packt, May 2026